Conditional probability

1: Set Notation

How sets are used to describe a sample space/event and how notation like $A \cap B$ is used to combine sets.

2: Conditional Probability in Venn Diagrams

The notation $P(A|B)$ means "the probability of A given that B happened". How we can find such probabilities using a Venn Diagram.

3: Formula for Conditional

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

4: Tree Diagrams

"I have 3 red and 4 green balls in a bag. I take one ball out the bag, keep it, then take another. **Given that** the second ball was green, determine the probability the first was red."

RECAP :: Using sets for sample spaces and events

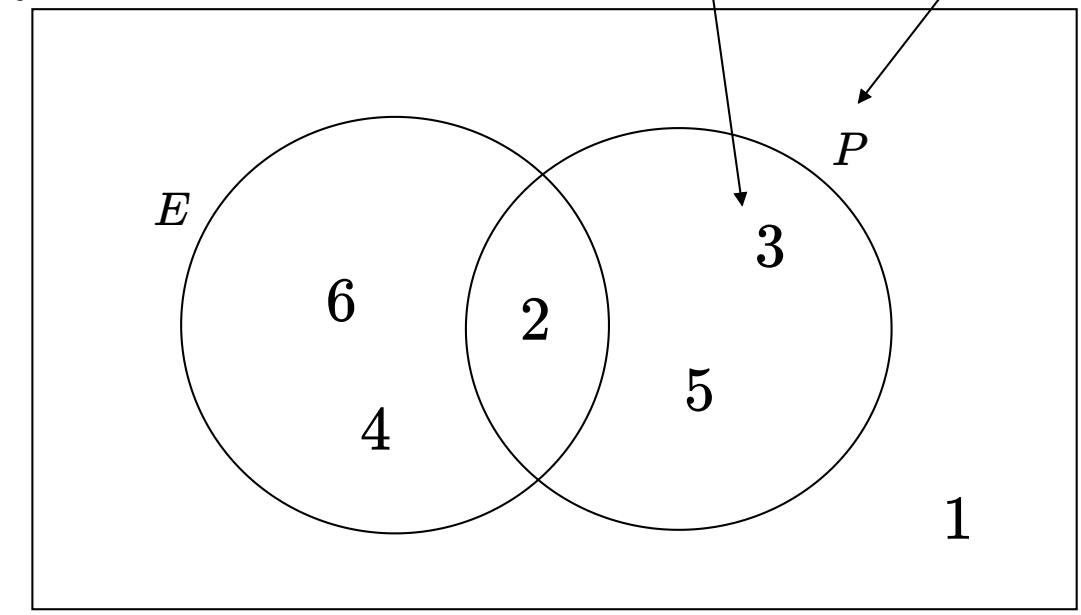
In general, sets are used to represent **collections of items**.

A **sample space** is **set of all possible outcomes**. We use ξ (Greek 'Xi'), or sometimes just S, to represent this set. We use a rectangle in a Venn Diagram.

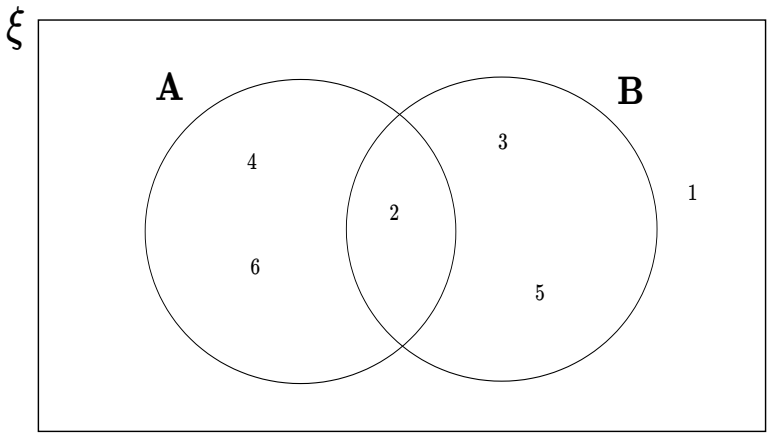
Each number represents an **outcome**.

In probability, an **event** is a **set of one or more outcomes**. These are the circles in the Venn Diagram. We use capital letters for the variables representing sets.

E = "rolling an even number"
P = "rolling a prime number"



Combining events/sets



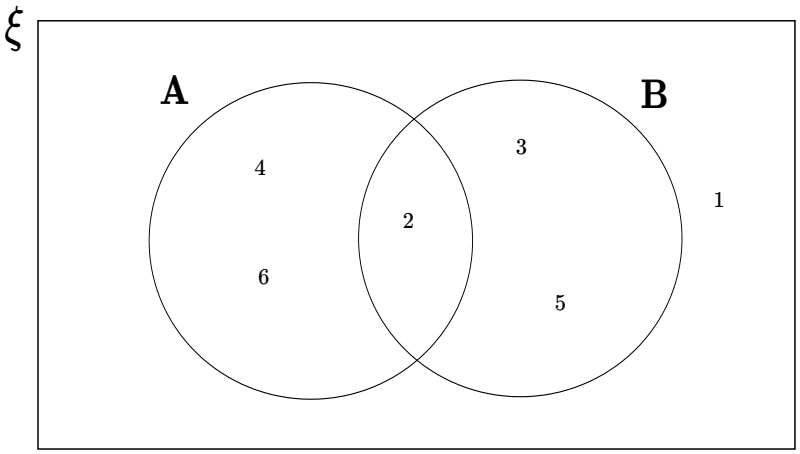
ξ = the whole sample space (1 to 6)

A = even number on a die thrown

B = prime number on a die thrown

	What does it mean in this context?	What is the resulting set of outcomes?
A'	?	?
$A \cup B$	?	?
$A \cap B$	?	?

Some fundamentals



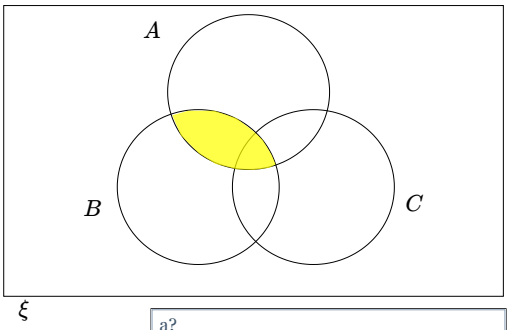
S = the whole sample space (1 to 6)

A = even number on a die thrown

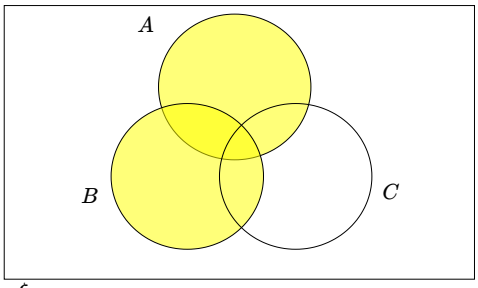
B = prime number on a die thrown

	What does it mean in this context?	What is the resulting set of outcomes?
$A \cap B\prime$	?	?
$(A \cup B)\prime$	?	?
$(A \cap B)'$	?	?

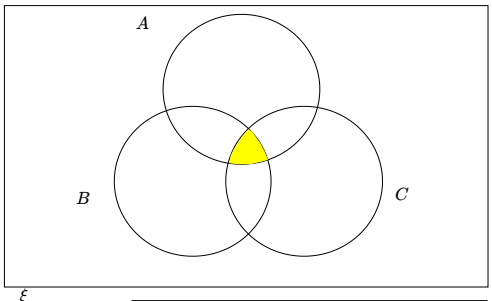
What area is indicated?



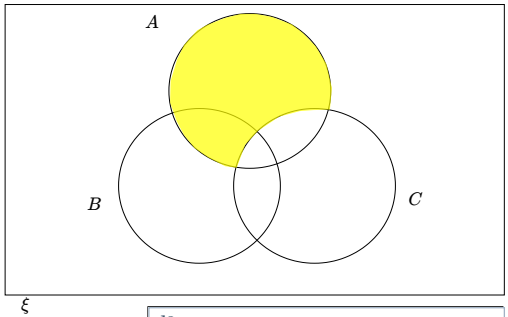
a?



b?

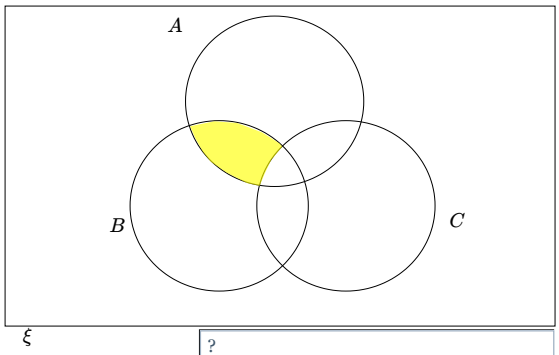


c?

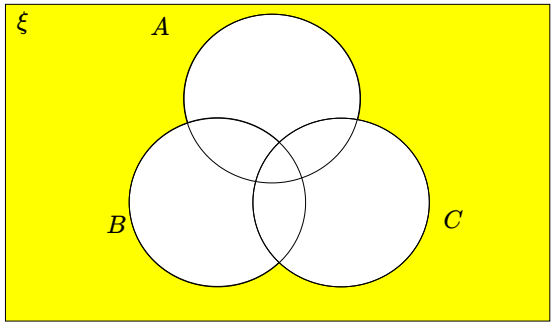


d?

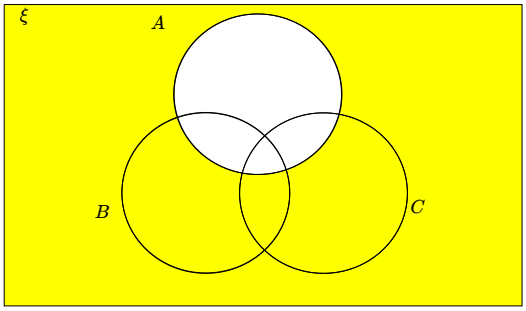
What area is indicated? 7



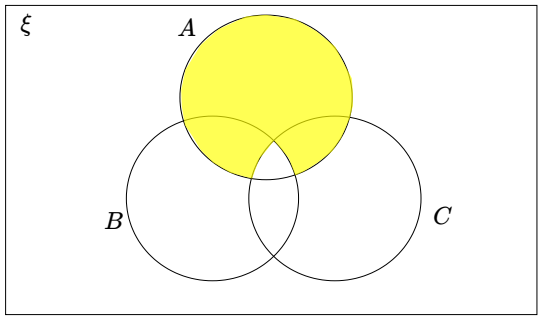
?



? or ?

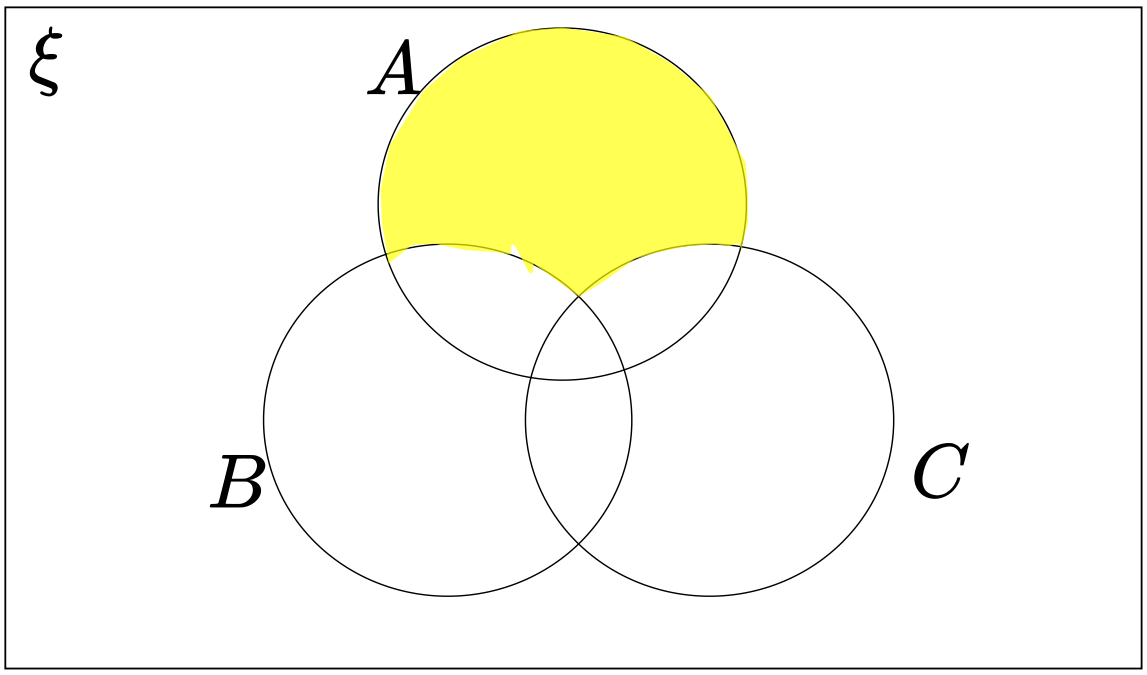


?



?

What area is indicated? 8



?

Venn Diagram can either contain:
(a) The **specific outcomes** in each set
(b) The number of items in the set (i.e. **frequencies**)
(c) The **probability** of being in that set.

This will usually be stated or made obvious from the context.

[Textbook] A card is selected at random from a pack of 52 playing cards. Let A be the event that the card is an ace and D the event that the card is a diamond. Find:
a) $P(A \cap D)$ b) $P(A \cup D)$ c) $P(A')$ d) $P(A' \cap D)$

? Venn Diagram

? a

? b

? c

? d

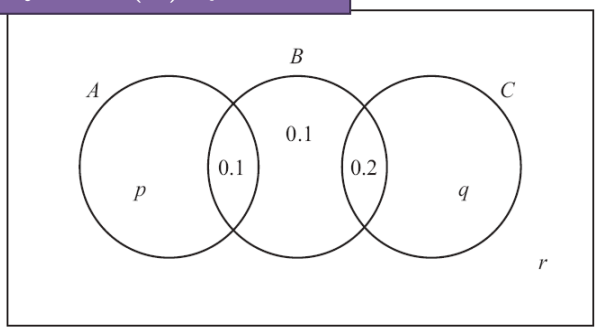
[Textbook] Given that $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cap B) = 0.25$,
a. Explain why events A and B are not independent.
Given also that $P(C) = 0.2$, that events A and C are mutually exclusive and that events B and C are independent,
b. Draw a Venn diagram to illustrate the events A, B and C, showing the probabilities for each region.
c. Find $P((A \cap B') \cup C)$

?

?

?

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(a) ?

The Venn diagram in Figure 1 shows three events A, B and C and the probabilities associated with each region of B. The constants p, q and r each represent probabilities associated with the three separate regions outside B.

The events A and B are independent.

(a) Find the value of p. **(3)**

9 A, B and C are three events with $P(A) = 0.55$, $P(B) = 0.35$ and $P(C) = 0.4$. $P(A \cap C) = 0.2$.
Given that A and B are mutually exclusive and B and C are independent,

a draw a Venn diagram to illustrate the probabilities. **(4 marks)**

b Find:

i $P(A' \cap B')$ **(1 mark)**

ii $P(A \cup (B \cap C'))$ **(1 mark)**

iii $P((A \cap C)' \cup B')$ **(1 mark)**

a?

b(i)?

b(ii)?

b(iii)?

10 A, B and C are three events with $P(A) = 0.25$, $P(B) = 0.4$, $P(C) = 0.45$ and $P(A \cap B \cap C) = 0.1$.
Given that A and B are independent, B and C are independent, and $A \cap B' \cap C = \emptyset$,

a draw a Venn diagram to illustrate the probabilities.

b Find:

i $P(A' \cap (B' \cup C))$

ii $P((A \cup B) \cap C)$

c State, with reasons, whether events A' and C independent.

a?

b?

c?

11 Members of a school book club read either murder mysteries (M), ghost stories (G) or epic fiction (E). $P(M) = 0.5$, $P(G) = 0.4$ and $P(E) = 0.6$. Given that no one reads both ghost stories and epic fiction and that $P(M \cap G) = 0.3$,

a draw a Venn diagram to illustrate these probabilities. **(4 marks)**

b Find:

i $P(M \cup G)$ **ii** $P((M \cap G) \cup (M \cap E))$ **(2 marks)**

c Are the events G' and M independent? You must justify your answer. **(2 marks)**

a?

b?

c?

12 Given that events A and B are independent and that $P(A) = x$ and $P(B) = y$, find, in terms of x and y:

a $P(A \cap B)$ **(2 marks)**

b $P(A \cup B)$ **(2 marks)**

c $P(A \cup B')$ **(2 marks)**

a?

b?

c?

EXTENSION (Challenge Page 20) 16

Given that events A, B and C are all independent and that $P(A) = x$, $P(B) = y$, $P(C) = z$, find, in terms of x, y and z:

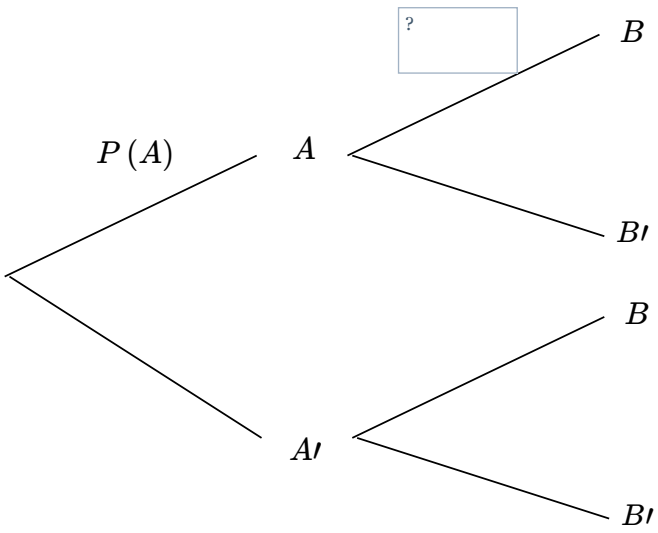
a $P(A \cap B \cap C)$ **b** $P(A \cup B \cup C)$ **c** $P((A \cup B') \cap C)$

a?

b?

c?

Think about how we formed a probability tree at GCSE:



$P(A \cap B)$
$= P($?

?

Memory Tip: You're dividing by the event you're conditioning on.

1 The following two-way table shows what foreign language students in Year 9 study.
B is the event that the student is a boy.
F is the event they chose French as their language.

	B	**B′**	Total
F	14	38	52
F′	26	22	48
Total	40	60	100

a Determine the probability of:
$P(F|B')$

?

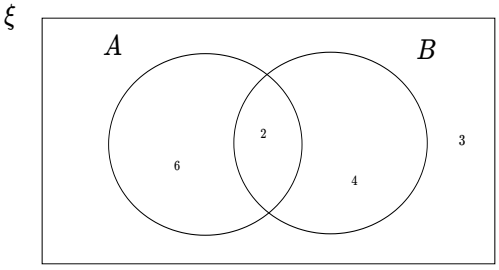
Method 2: Restricted sample space.

?

b $P(B|F')$?

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2 Using the Venn Diagram, determine:



a $P(A|B)$

?

Method 2: Restricted sample

?

b $P(A'|B') =$?

c $P(B|A \cup$? $= \frac{1}{2}$

Further Examples

a Given that $P(A) = 0.5$ and $P(A \cap B) = 0.3$, what is $P(B|A)$

?

Note: The 'restricted sample space' method also works for Venn Diagrams with probabilities.

b Given that $P(Y) = 0.6$ and $P(X \cap Y) = 0.4$, what is $P(X'|Y)$?

(Hint: Drawing a Venn Diagram will help!)

?

$\frac{0.2}{0.6} = \frac{1}{3}$

c Given that $P(A) = 0.5$, $P(B) = 0.5$ and $P(A \cap B) = 0.4$, what is $P(B|A')$

?

Check your understanding

The events E and F are such that

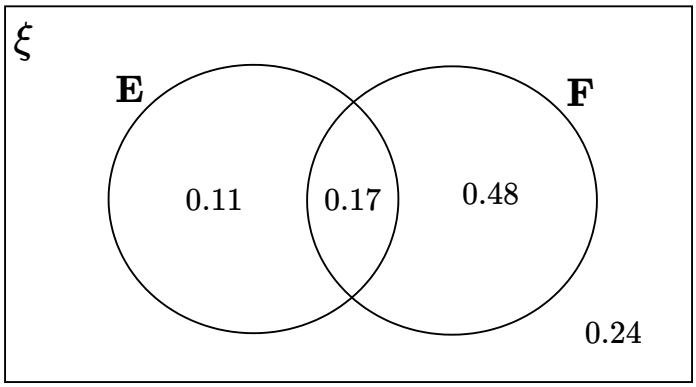
$P(E) = 0.28 \quad P(E \cup F) = 0.76 \quad P(E \cap F') = 0.11$

Find

a) $P(E \cap F) =$? 7

b) $P(F) =$?

c) $P(E'|F') =$? $\frac{24}{35}$



1 $P(A \cap B') = 0.4, P(A \cup B) = 0.75$

Then:

$P(B) =$?

$P(A' \cap B') =$?

2 $P(A) = 0.47$ and $P(A \cap B) = 0.12$ and $P(A' \cap B') = 0.03$

Then:

$P(A|B')$?

$\mathbf{P(B')}$ $\mathbf{0.38}$ $\mathbf{38}$

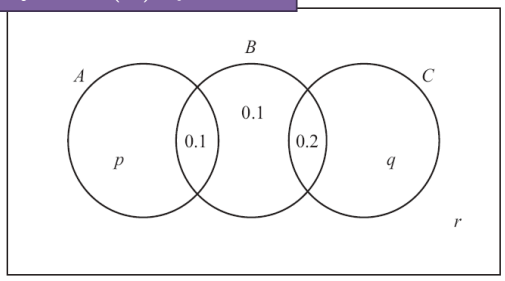
3 $P(A') = 0.7, P(B') = 0.2, P(A \cap B') = 0.1$

Then:

$P(A \cup B') = \mathbf{P(A)}$?

$P(B|A') = \frac{\mathbf{P(A' \cap B}}{\mathbf{P(A')}}$?

May 2013 (R) Q6



The Venn diagram in Figure 1 shows three events A, B and C and the probabilities associated with each region of B. The constants p, q and r each represent probabilities associated with the three separate regions outside B.

The events A and B are independent.

(a) Find the value of p. **(3)**

Given that $P(B|C) = \frac{5}{11}$,

(b) find the value of q and the value of r (4)

(c) Find $P(A \cup C|B)$ (2)

(a) **(From earlier)**

$0.1 = (p + 0.1) \times 0.4$

$p + 0.1 = 0.25$

$p = 0.15$

(b) ?

4

(c) ?

8 120 students are asked about their viewing habits. 56 say they watch sports (S) and 77 say they watch dramas (D). Of those who watch dramas, 18 also watch sports.

a Draw a two-way table to show this information. **(2 marks)**

b One student is chosen at random. Find:

i $P(D')$ **(1 mark)**

ii $P(S' \cap D')$ **(1 mark)**

iii $P(S|D)$ **(1 mark)**

iv $P(D'|S)$ **(1 mark)**

a?

b?

9 A rambling group is made up of 63 women and 47 men. 26 of the women and 18 of the men use a walking stick.

a Draw a two-way table to show this information. **(2 marks)**

b One rambler is chosen at random. Find:

i P(Uses a stick) **(1 mark)**

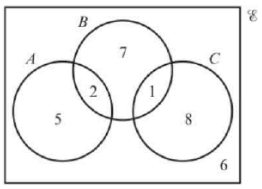
ii P(Uses a stick|Female) **(1 mark)**

iii P(Male|Uses a stick) **(1 mark)**

a?

b?

7 The Venn diagram shows the number of students in a class who watch any of 3 popular TV programmes A, B and C.



One of these students is selected at random. Given that the student watches at least one of the TV programmes, find the probability that the student watches:

a programme C **(2 marks)**

b exactly two of the programmes. **(2 marks)**

c Determine whether or not watching programme B and watching programme C are statistically independent. **(3 marks)**

a?

b?

c?

8 Three events, A, B and C are such that A and B are mutually exclusive and B and C are independent. $P(A) = 0.2$, $P(B) = 0.6$ and $P(C) = 0.5$. Given that $P(A' \cap B' \cap C') = 0.1$,

a draw a Venn diagram to show the probabilities for events A, B and C. **(4 marks)**

b Find:

i $P(A|C)$ **(1 mark)**

ii $P(B|C')$ **(1 mark)**

iii $P(C|(A \cup B))$ **(1 mark)**

a?

b?

9 A doctor completes a medical study of 100 people, 5 of whom are known to have an illness and 95 of whom are known not to. A diagnostic test is applied. All 5 of the people with the illness test positive, and 10 people without the illness also test positive. Given that event A = person has the disease and event B = person tests positive,

a draw a Venn diagram to represent this situation. **(3 marks)**

b Calculate $P(A|B)$. **(2 marks)**

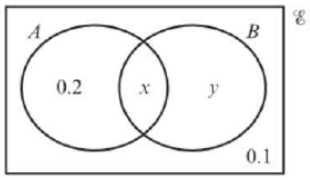
c With reference to your answer to part **b**, comment on the usefulness of the diagnostic test. **(2 marks)**

a?

b?

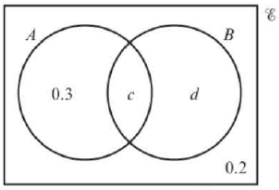
c?

11 The Venn diagram shows the probabilities for two events, A and B. Given that $P(A|B) = P(B')$, find the values of x and y.



?

12 The Venn diagram shows the probabilities for two events, A and B. Given that $P(A|B) = P(A')$, find the values of c and d.



?

[Classic puzzle] I have 2 children. One of them is a boy. What's the probability the other is a boy?

?

If events A and B are independent.

$P(A \cap B) =$?

$P(A \mid B) =$?

If events A and B are mutually exclusive:

$P(A \cap B) =$?

$P(A \cup B) =$?

In general:

$P(A \mid B) =$?

We first encountered this in the previous section.

$P(A \cup B) =$?

Edexcel S1

6. Explain what you understand by
(a) a sample space, **(1)**
(b) an event. **(1)**

Two events A and B are independent, such that $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{4}$

Find
(c) $P(A \cap B)$, **(1)**
(d) $P(A \mid B)$, **(2)**
(e) $P(A \cup B)$. **(2)**

a) ?
b) ?
c) ?
d) ?
e) ?

[Textbook] C and D are two events such that $P(C) = 0.2$, $P(D) = 0.6$ and $P(C|D) = 0.3$. Find:
a. $P(C \cap D)$ b. $P(D|C)$ c. $P(C \cup D)$

a?
b?
c?

Hints for (b): You saw the words "are independent". So write out $P(A \cap B) = P(A)\,P(B)$.
Also, you're given $P(A \cup B)$ which suggests you might be able to use the Addition Rule.

a?
b?
c?
d?

10. [Jan 2012 Q2] (a) State in words the relationship between two events R and S when $P(R \cap S) = 0$. **(1)**

The events A and B are independent with $P(A) = \frac{1}{4}$ and $P(A \cup B) = \frac{2}{3}$. Find
(b) $P(B)$, **(4)**
(c) $P(A' \cap B)$, **(2)**
(d) $P(B'|A)$. **(2)**

Edexcel S1

9. Three events A, B and C are defined in the sample space S. The events A and B are mutually exclusive and A and C are independent.

(a) Draw a Venn diagram to illustrate the relationships between the 3 events and the sample space. **(3)**

Given that $P(A) = 0.2$, $P(B) = 0.4$ and $P(A \cup C) = 0.7$, find

(b) $P(A \mid C)$, **(2)**

(c) $P(A \cup B)$, **(2)**

(d) $P(C)$. **(4)**

a) ?

b) ?

c) ?

d) ?

P

0.7

0.5

P

If I were to identify two tips that will possible help you the most in probability questions:

If you see the words 'given that', Immediately write out the law for conditional probability.

Example: "Given Bob walks to school, find the probability that he's not late. . ."

First thing you should write: ? ...

If you see the words 'are independent', Immediately write out the laws for independence.
(Even before you've finished reading the question!)

Example: "A is independent from B. . ."

First thing you should write: $P($?

$P(A|B) = P(A)$

If you're stuck on a question where you have to find a probability given others, it's probably because you've failed to take into account that two events are independent or mutually exclusive, or you need to use the conditional probability or additional law.

9 Given that $P(A) = 0.42$, $P(B) = 0.37$ and $P(A \cap B) = 0.12$. Find:

a $P(A \cup B)$ **(2 marks)**

b $P(A \mid B')$ **(2 marks)**

The event C has $P(C) = 0.3$.

The events B and C are mutually exclusive and the events A and C are independent.

c Find $P(A \cap C)$. **(2 marks)**

d Draw a Venn diagram to illustrate the events A, B and C, giving the probabilities for each region. **(4 marks)**

e Find $P((A' \cup C)')$. **(2 marks)**

a?

b?

c?

d?

e?

10 Three events A, B and C are such that $P(A) = 0.4$, $P(B) = 0.7$, $P(C) = 0.4$ and $P(A \cap B) = 0.3$. Given that A and C are mutually exclusive and that B and C are independent, find:

a $P(B \cap C)$ **(1 mark)**

b $P(B|C)$ **(1 mark)**

c $P(A|B')$ **(1 mark)**

d $P((B \cap C)|A')$ **(1 mark)**

a?

b?

c?

d?

e?

11 Anna and Bella are sometimes late for school. The events A and B are defined as follows:
A is the event that Anna is late for school
B is the event that Bella is late for school
$P(A) = 0.3$, $P(B) = 0.7$ and $P(A' \cap B') = 0.1$. On a randomly selected day, find the probability that:

a both Anna and Bella are late to school **(1 mark)**

b Anna is late to school given that Bella is late to school. **(2 marks)**

Their teacher suspects that Anna and Bella being late for school is linked in some way.

c Comment on his suspicion, showing your working. **(2 marks)**

a?

b?

c?

12 John and Kayleigh play darts in the same team. The events J and K are defined as follows:
J is the event that John wins his match
K is the event that Kayleigh wins her match
$P(J) = 0.6$, $P(K) = 0.7$ and $P(J \cup K) = 0.8$.
Find the probability that:

a both John and Kayleigh win their matches **(1 mark)**

b John wins his match given that Kayleigh loses hers **(2 marks)**

c Kayleigh wins her match given that John wins his. **(2 marks)**

d Determine whether the events J and K are statistically independent.
You must show all your working. **(2 marks)**

a?

b?

c?

d?

The discrete random variable X has a probability function $P(X = x) = kx, x = 1, 2, 3, 4$. Find:

a the value of k **b** $P(x = 5|X > 2)$ **c** $P(X \text{is odd}|\ X \text{ is prime})$

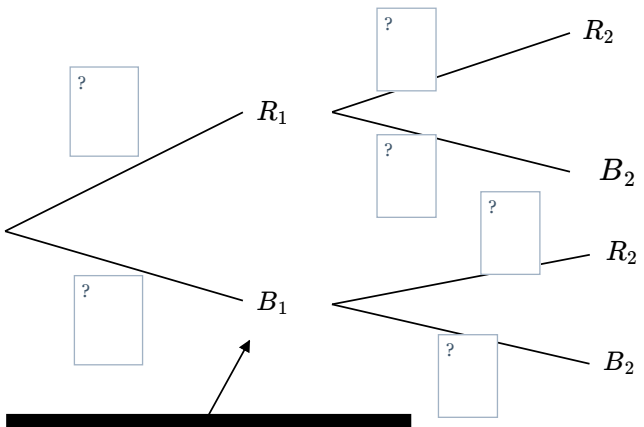
a?

b?

c?

Probability Trees can be used to **determine a conditional probability.**

Example: You have two bags, the first with 5 red balls and 5 blue balls, and the second with 3 red balls and 6 blue balls. You first pick a ball from the first bag, and place it in the second. You then pick a ball from the second bag. Complete the tree diagram.



Note: Use variable subscripting to indicate what pick you're referring to.

Hence find the probability that:

a) **You pick a red ball on your second pick.**

$P(R_2) = P(R_1 \cap R_2) + P(B_1 \cap R_2) = \frac{1}{5} + \frac{3}{20} = \frac{7}{20}$

b) **Given that your second pick was red, the first pick was also red.**

?

?

Edexcel S1 May 2009 Q2

On a randomly chosen day the probability that Bill travels to school by car, by bicycle or on foot is $\frac{1}{2}, \frac{1}{6}$ and $\frac{1}{3}$ respectively. The probability of being late when using these methods of travel is $\frac{1}{5}, \frac{2}{5}$ and $\frac{1}{10}$ respectively.

Given that Bill is late, find the probability that he did not travel on foot. **(4)**

a) draw a probability Tree

?

B) Calculate the probability

?

Edexcel S1

6. [Jan 2006 Q4] A bag contains 9 blue balls and 3 red balls. A ball is selected at random from the bag and its colour is recorded. The ball is not replaced. A second ball is selected at random and its colour is recorded.

(*a*) Draw a tree diagram to represent the information. **(3)**

Find the probability that

(*a*) the second ball selected is red, **(2)**

(*b*) both balls selected are red, given that the second ball selected is red. **(2)**

(a) ?

(b) ?

(c) ?

6 A bag contains 4 blue balls and 7 green balls. A ball is selected at random from the bag and its colour is recorded. The ball is not replaced. A second ball is selected at random and its colour is recorded.

a Draw a tree diagram to represent the information. **(3 marks)**

Find the probability that:

b the second ball selected is green **(2 marks)**

c both balls selected are green, given that the second ball selected is green. **(2 marks)**

a?

b?

c?

7 In an engineering company, factories A, B and C are all producing tin sheets of the same type. Factory A produces 25% of the sheets, factory B produces 45% and the rest are produced by factory C. Factories A, B and C produce flawed sheets with probabilities 0.02, 0.07 and 0.04 respectively.

a Draw a tree diagram to represent this information. **(3 marks)**

b Find the probability that a randomly selected sheet is:

i produced by factory B and flawed **(2 marks)**

ii flawed. **(3 marks)**

c Given that a randomly selected sheet is flawed, find the probability that it was produced by factory A. **(3 marks)**

a?

b?

c?

d?

8 A genetic condition is known to be present in 4% of a population. A test is developed to help determine whether or not someone has the genetic condition.
If a person has the condition, the test is positive with probability 0.9.
If a person does not have the condition, the test is positive with probability 0.02.

a Draw a tree diagram to represent this information. **(3 marks)**

A person is selected at random from the population and tested for this condition.

b Find the probability that the test is negative. **(3 marks)**

A doctor randomly selects a person from the population and tests them for the condition.

Given that the test is negative,

c find the probability that they do have the condition. **(2 marks)**

d Comment on the effectiveness of this test. **(1 mark)**

a?

b?

c?

d?

9 On a randomly chosen day the probabilities that Bill travels to work by car, by bus or by train are 0.1, 0.6 and 0.3 respectively. The probabilities of being late when using these methods of travel are 0.55, 0.3 and 0.05 respectively.

a Draw a tree diagram to represent this information. **(3 marks)**

b Find the probability that on a randomly chosen day,

i Bill travels by train and is late **(2 marks)**

ii Bill is late. **(2 marks)**

c Given that Bill is late, find the probability that he did not travel by car. **(4 marks)**

a?

b?

c?

11 A box of jelly beans contains 7 sweet flavours and 3 sour flavours. Two of the jelly beans are taken one after the other and eaten. Emilia wants to find the probability that both jelly beans eaten are sweet given that at least one of them is. Her solution is shown below:

P(both jelly beans are sweet) $= \frac{7}{10} \times \frac{7}{10} = \frac{49}{100}$

P(at least one jelly bean is sweet)

$= 1 - \text{P(neither are sweet)} = 1 - \frac{3}{10} \times \frac{3}{10} = \frac{91}{100}$

P(both are sweet given at least one is sweet)

$= \frac{\frac{49}{100}}{\frac{91}{100}} = \frac{49}{91}$

Identify Emilia's mistake and find the correct probability. **(4 marks)**

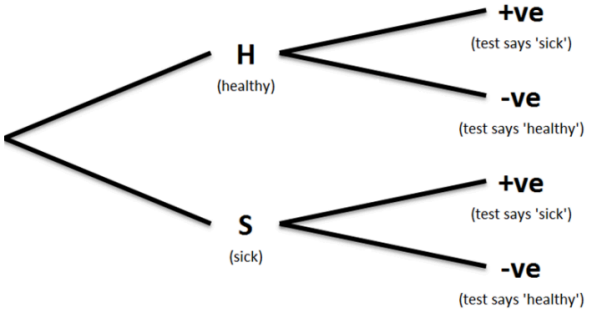
?

Almost all medical tests have a small chance of a 'false positive' or a 'false negative'.

- **False positive**: The test says you **do** have the condition when you really **don't**. (you're fine, but the test thinks you're sick)
- **False negative**: The test says you **don't** have the condition when you really **do**. (you're sick, but the test thinks you're fine)

- The test for a particular disease has a false positive rate of 5%.
  *This means that 5% of people who **don't** have the disease will be told that they **do**.*
- The test has a false negative rate of 1%.
  *This means that 1% of people who **do** have the disease will be told that they **don't**.*

Assuming this particular disease affects 10%of people, complete the tree diagram below:



Use the tree diagram to calculate the probabilities for each of the four outcomes:

	Healthy	Sick
Tested Positive		
Tested Negative		

You're sick, and the test was correct

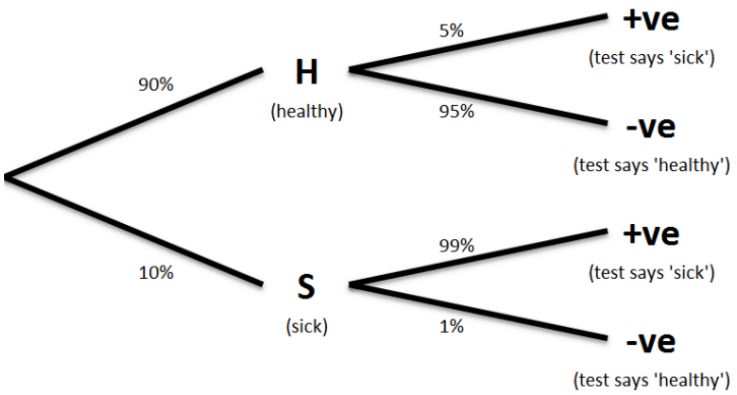
You're sick, but the test thinks you're fine
FALSE NEGATIVE:
You may not get the treatment you need.

You're fine, but the test thinks you're sick
FALSE POSITIVE:
You may get treatment you don't need, or be worried for no reason.

You're fine, and the test was correct

If the test says I'm sick, what is the chance that I really am?

Hint: Imagine 1000 people take the test. Work out how many would test positive, and out of those people, how many are really sick?



$0.9 \times 0.05 = \mathbf{0.045}$ $0.9 \times 0.95 = \mathbf{0.855}$ $0.1 \times 0.99 = \mathbf{9.9}$ $0.1 \times 0.01 = \mathbf{0.001}$

	Healthy	Sick
Tested Positive	4.5%	9.9%
Tested Negative	85.5%	0.1%

You're sick, and the test was correct

You're sick, but the test thinks you're fine
FALSE NEGATIVE:
You may not get the treatment you need.

You're fine, but the test thinks you're sick
FALSE POSITIVE:
You may get treatment you don't need, or be worried for no reason.

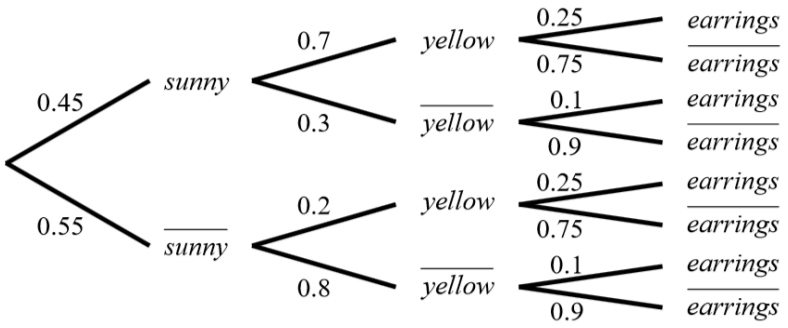
You're fine, and the test was correct

If the test says I'm sick, what is the chance that I really am?

For every 1000 *people,* $45 + 99 = 144$ *of the population will test positive.*
Of that 144,99 *are genuinely sick.*
$\frac{99}{144} = 0.6875 = \mathbf{68.75\%}$ ***chance that you're really sick.***

Mary chooses her outfit based on the weather. The probability that it will be sunny when Mary wakes up is 0.45. The probability that Mary will wear something yellow is 0.7 when it is sunny, or 0.2 if it isn't. If Mary wears yellow, there is a 25% chance that she will wear her favourite earrings, but only a 10% chance if she is not wearing yellow.

What is the probability that Mary will be wearing yellow today, given that she is not wearing her favourite earrings?



Probability of event A, given event B: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$

$$P(A \cap B) = P(sunny, yellow, \overline{earrings}) + P(\overline{sunny}, yellow, \overline{earrings})$$
$$= 0.23625 + 0.0825$$
$$= 0.31875$$

$$P(B) = P(\overline{earrings})$$
$$= 0.23625 + 0.1215 + 0.0825 + 0.396$$
$$= 0.83625$$

$$P(yellow \mid \overline{earrings}) = \frac{0.31875}{0.83625}$$

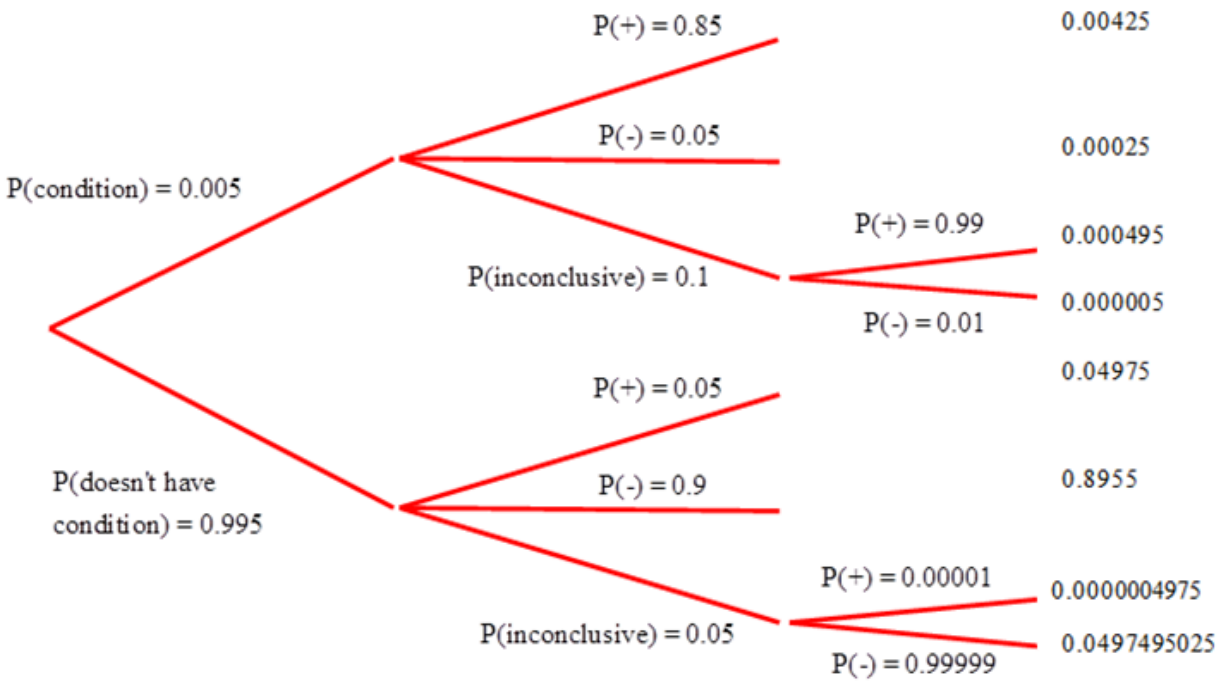
$$= 0.38116...$$
$$= 0.381 (3 s.f.)$$

A test for whether a patient has a particular condition has been trialled extensively by the pharmaceutical companies manufacturing the test. They find that 0.5% of the population have this particular medical condition. After testing, the outcomes are positive, negative or inconclusive.

It is found that if the patient has the condition then there is an 85% chance that they test positive, a 5% chance that they test negative or a 10% chance that the results are inconclusive. If the patient doesn't have the condition then there is a 90% chance that they test negative, a 5% chance that they test positive and a 5% chance that the tests are inconclusive.

If the results are inconclusive then the patient has to have a thorough examination. If they do not have the condition then the probability that a thorough examination determines they have the condition is 0.001%. If the patient does have the condition then the probability that a thorough examination detects this is 99%.

What is the probability that given they don't have the condition they test positive?



$$P(\text{Test positive} \mid \text{Don't have condition}) = \frac{P(\text{Test positive} \cap \text{Don't have condition})}{P(\text{Don't have condition})}$$

$$= \frac{0.04975 + 0.0000004975}{0.995} = 0.0500005$$